MSG No. 107.228 $I4mm1'$ [Type II, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `4mm1'`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,4,5,6,7,8,17,18,19,20,21,22,23,24]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,10,11,12,13,14,15,16,25,26,27,28,29,30,31,32]`

Table 2: Wyckoff site: `4b`, site symmetry: `2mm.1'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,4,5,6,17,20,21,22]`
2	$\left[\frac{1}{2},\ 0,\ z\right]$	`[2,3,7,8,18,19,23,24]`
3	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[9,12,13,14,25,28,29,30]`
4	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[10,11,15,16,26,27,31,32]`

Table 3: Wyckoff site: `8c`, site symmetry: `..m1'`

No.	position	mapping
1	$[x,\ x,\ z]$	`[1,8,17,24]`
2	$[-x,\ x,\ z]$	`[2,5,18,21]`
3	$[x,\ -x,\ z]$	`[3,6,19,22]`
4	$[-x,\ -x,\ z]$	`[4,7,20,23]`
5	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,16,25,32]`
6	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[10,13,26,29]`
7	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[11,14,27,30]`
8	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[12,15,28,31]`

Table 4: Wyckoff site: `8d`, site symmetry: `.m.1'`

No.	position	mapping
1	$[x,\ 0,\ z]$	`[1,6,17,22]`
2	$[0,\ x,\ z]$	`[2,8,18,24]`
3	$[0,\ -x,\ z]$	`[3,7,19,23]`
4	$[-x,\ 0,\ z]$	`[4,5,20,21]`
5	$\left[x+\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,14,25,30]`
6	$\left[\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[10,16,26,32]`
7	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[11,15,27,31]`
8	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[12,13,28,29]`

Table 5: Wyckoff site: `16e`, site symmetry: `11'`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1,17]`
2	$[-y,\ x,\ z]$	`[2,18]`

continued ...

Table 5

No.	position	mapping
3	$[y,\,-x,\,z]$	[3,19]
4	$[-x,\,-y,\,z]$	[4,20]
5	$[-x,\,y,\,z]$	[5,21]
6	$[x,\,-y,\,z]$	[6,22]
7	$[-y,\,-x,\,z]$	[7,23]
8	$[y,\,x,\,z]$	[8,24]
9	$\left[x+\frac{1}{2},\,y+\frac{1}{2},\,z+\frac{1}{2}\right]$	[9,25]
10	$\left[\frac{1}{2}-y,\,x+\frac{1}{2},\,z+\frac{1}{2}\right]$	[10,26]
11	$\left[y+\frac{1}{2},\,\frac{1}{2}-x,\,z+\frac{1}{2}\right]$	[11,27]
12	$\left[\frac{1}{2}-x,\,\frac{1}{2}-y,\,z+\frac{1}{2}\right]$	[12,28]
13	$\left[\frac{1}{2}-x,\,y+\frac{1}{2},\,z+\frac{1}{2}\right]$	[13,29]
14	$\left[x+\frac{1}{2},\,\frac{1}{2}-y,\,z+\frac{1}{2}\right]$	[14,30]
15	$\left[\frac{1}{2}-y,\,\frac{1}{2}-x,\,z+\frac{1}{2}\right]$	[15,31]
16	$\left[y+\frac{1}{2},\,x+\frac{1}{2},\,z+\frac{1}{2}\right]$	[16,32]